Mortality Patterns: Retrospective Estimation and scenario-based forecasts- **GitHub Documentation**

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Introduction & Overall Description

The GitHub repository 'mortality_estimation' has been created to examine mortality rates in different countries. The main objectives of this study are to:

- Project or **estimate the crude death** rate using measured values of various predictor variables,
- **forecast scenarios** for the next five months based on the predictions of these predictor variables.

To achieve these two main objectives, the GitHub repository is organised into **10** separate folders, as shown in Figure 1. The following section will first guide you through the libraries and versions of R required for successful execution of the different steps. After that, we'll dive into a detailed exploration of the different folders, most of which are made up of four main parts: data, results, visualisation and code.

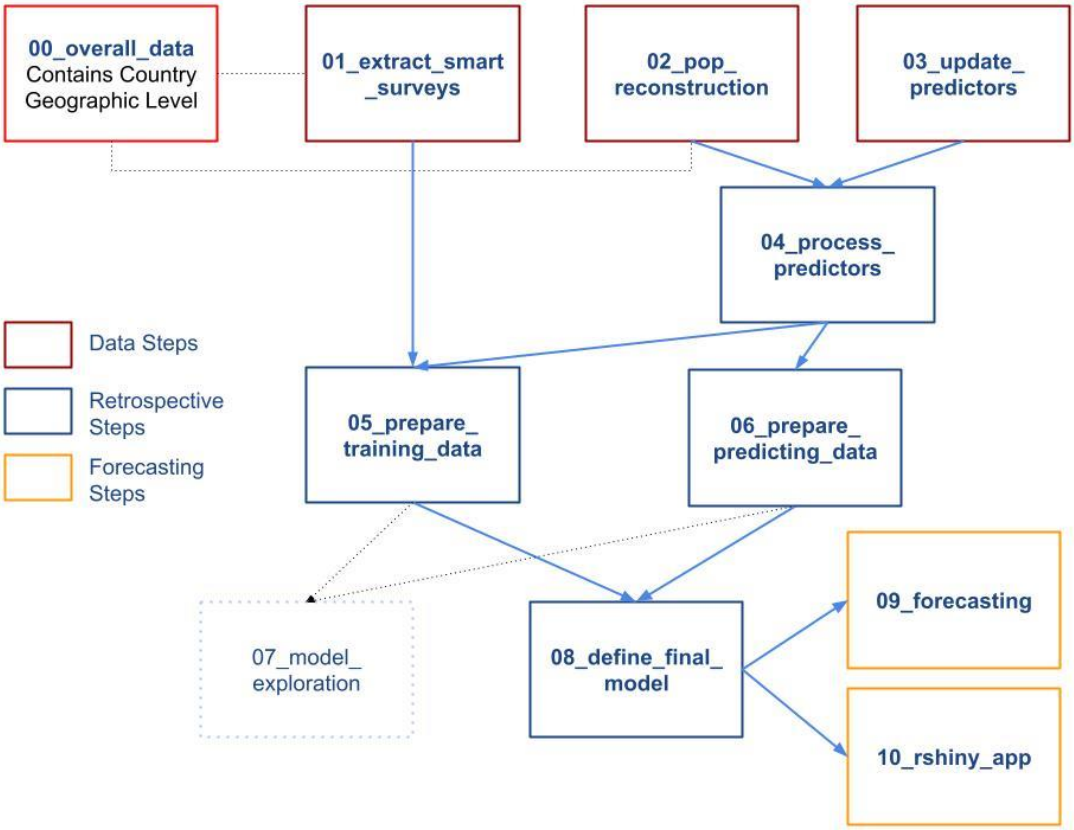

Figure 1. Visualization of the Github Structure

Note: If you only wish to view and experiment with the **RShiny app** for forecasting scenarios, please refer to the next section (Packages, libraries and versions) to ensure that you have the necessary packages to launch the RShiny app. Then go to the folder entitled "10_rshiny_app" and follow the instructions provided in the following video:

https://lshtm.cloud.panopto.eu/Panopto/Pages/Viewer.aspx?id=7ea184cf-2028-4554-867a-b0d901059162

Packages, Libraries & Versions

To run the RShiny code successfully, make sure you have access to the latest version of RStudio (R version 4.2.2). Next, download this GitHub repository to your local device and double-click (or open) the 'MortalityEstimation.Rproj' file using RStudio.

After completing the above steps, download the required packages and load the necessary libraries. To do this, open and run the file entitled 'First_Run_Packages_Libraries.R'. Once you have run this file, you will be ready to launch all the code provided and the RShiny application.

It's important to **note** that we are utilizing the glmmTMB libraries, and the latest version may conflict with the most recent version of the Matrix package. Therefore, ensure that you have version 1.6-4 of the Matrix package. If you have any uncertainties, run the following R code line in the RStudio Console: `install.packages("Matrix", dependencies = TRUE, type='source')`

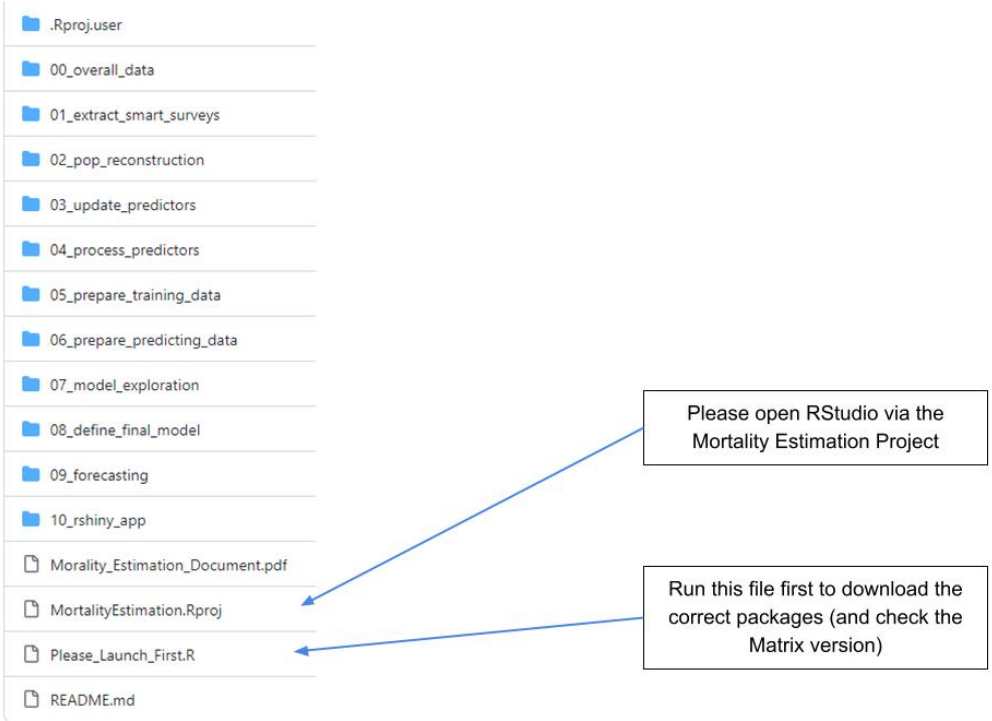

Figure 2. Steps to open the Project and Upload the right packages.

Mortality Estimation Steps

Step 1: Data Steps

Step 2: Retrospective Steps

Step3 : Forecasting Steps

RShiny App